

Intertidal survey of Kellow Sands and Thrimby Point

Report BEP/SR/2026/11 · Survey of 19 to 23 January 2026 · Issued February 2026

Summary

The four fixed transects at Kellow Sands and Thrimby Point were resurveyed in January, in the third winter since sediment recharge began. This report sets out the method, the profiles measured, and the change on each transect since the surveys began.

Levels were taken at five-metre intervals along each transect with a total station from the survey marks on the training wall and at the lifeboat house. Sediment samples were taken at the mid-shore station on each transect for particle size, and are reported by the laboratory separately.

Conditions during the survey were settled, with no swell and a spring range near the mean. Nothing about the week's weather is likely to have affected the levels measured.

Background

The Partnership has surveyed the intertidal at Kellow Sands since 2019, when the four transects were established to provide a baseline for the sediment recharge then being planned. The first two winters were surveyed before any material was placed. Placement began in the winter of 2023/24 and has been carried out in each winter since, most recently between November 2025 and January 2026.

The purpose of the survey is a narrow one. It is meant to answer whether the intertidal profile at Kellow Sands is gaining or losing sediment, and at what rate, on lines whose position is fixed. It is not a study of where the placed material goes, which would need tracer work the Partnership has not commissioned, and it cannot by itself attribute a change to the recharge rather than to the weather.

Method

Levels were taken with a total station from the two survey marks used since 2019, at five-metre intervals along each transect and at every break of slope between them. Each line was run twice, out and back, and the two sets agreed everywhere within the tolerance the method allows. Where they did not, which happened once on transect three, the point was re-observed.

The horizontal position of each line is fixed by a mark set in concrete at the landward end and by a pin in the training wall or the marsh edge at the seaward end. All eight marks were found and checked against the 2019 coordinates before work began. The pin at the seaward end of transect two has moved and was re-established from the landward mark; the correction is recorded in the level book and is smaller than the interval between observations.

Sediment samples were taken at the mid-shore station on each line, in the order in which the lines were surveyed, and delivered to the laboratory the same afternoon. Particle size is reported separately and is not discussed here.

The transects

Transect one runs from the seawall opposite the boatyard to the low water mark, crossing the widest part of the sand. It is the longest of the four lines and the only one on which the whole intertidal profile can be surveyed within a single tide.

Transects two and three are the pair the recharge was expected to affect. Both cross the placement footprint at its northern end, transect two some sixty metres from the discharge point and transect three about a hundred and forty. Neither has been moved since it was established, and the marks at both ends of both lines were found in good order.

The plan below shows the four lines as they were surveyed in January, together with the saltmarsh edge as it stood at the 1998 survey. The 1998 edge is drawn because it is the last edge surveyed on the ground rather than taken from photography, and because it is the line against which the advisers have asked that marsh extent be reported.

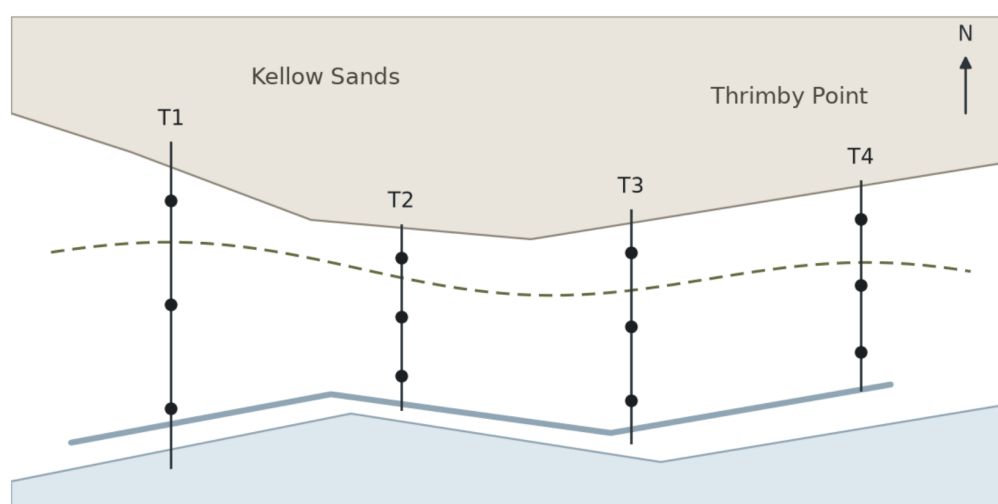


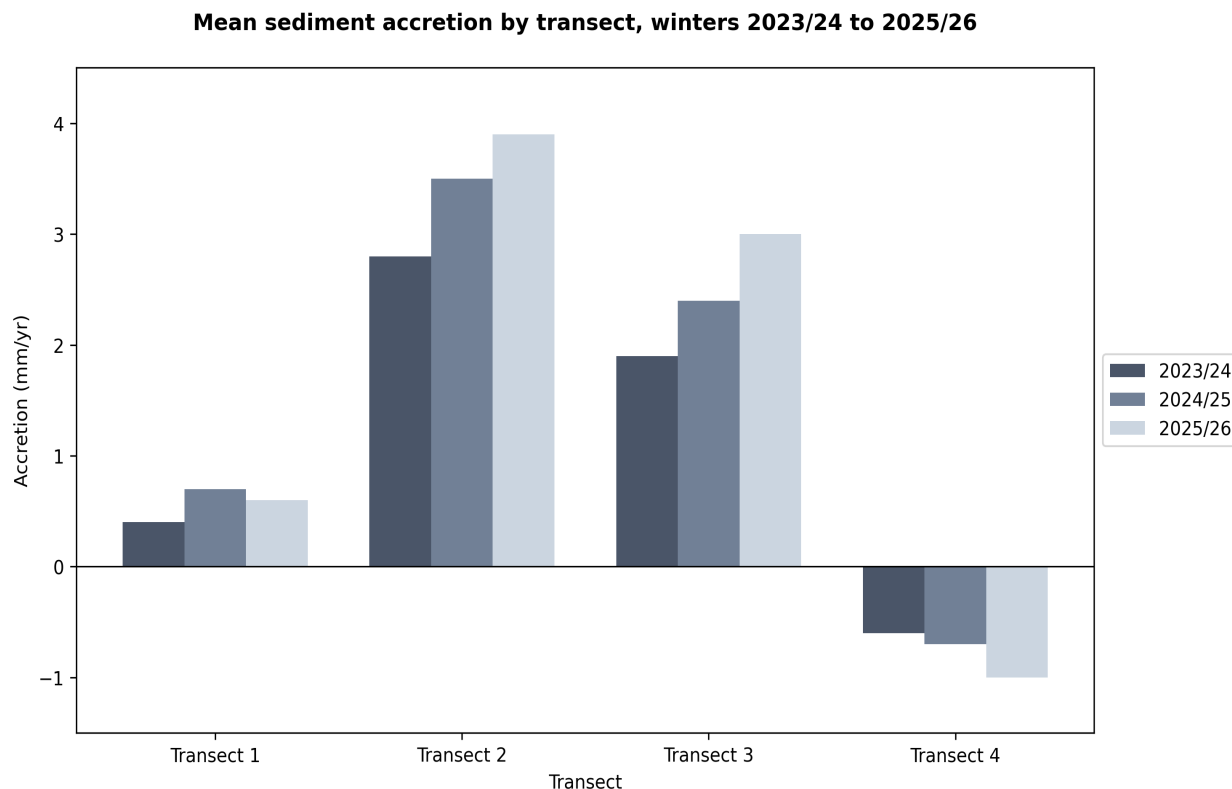
Figure 1. Survey transects at Kellow Sands and Thrimby Point. Sampling stations are shown as filled circles and the 1998 saltmarsh edge as a broken line.

Transect four crosses the toe of the training wall and has always behaved differently from the other three. It is retained because the wall itself is the reason for the difference, and because the Partnership was asked in 2019 to keep a line there.

The stations shown on the plan are those at which sediment samples are taken, and not the points at which levels were observed: the level observations are too close together to plot at this scale. A full set of profiles is bound into the copy of this report held by the secretariat.

Change since the first survey

Change is reported as a mean annual rate over each winter, taken across the whole intertidal part of the line. Accretion varied markedly between transects, and the pattern is the same in each of the three winters measured.



The rates plotted are means over the intertidal part of each line, weighted by the length of profile each observation represents. A mean of that kind conceals a good deal. On transect two most of the gain is in the upper third of the profile, and on transect four the loss is almost all at the toe of the wall; the profiles show this plainly and a mean does not.

The advisers' view, which the secretariat shares, is that three winters is too short a run to separate the effect of placement from the variation the estuary shows anyway. The transects should be resurveyed next January on the same lines and by the same method before anything is concluded from them.

The next survey

The profiles themselves, the level books and the mark check sheets are held by the secretariat and are available to Partnership members on request.

The next survey is due in January 2027. The secretariat proposes to keep the January window, and to ask the laboratory for particle size on the same four stations so that the series stays comparable.